Zilu Zhou (Max)

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Egocentric Computer Vision | MS Computer Science, Northeastern University

Research Interests

Egocentric computer vision: hand-centric perception in multi-person settings, multi-wearer video understanding in cooperative activity, and JEPA-style world models for anticipation, toward perception that supports real-time AR assistance.

Education

Northeastern University — Boston, MA
Master of Science in Computer Science, GPA: 3.5/4.0 — *Sep 2024 – Dec 2026*

University of California, Santa Barbara — Santa Barbara, CA
Bachelor in Applied Mathematics — *Sep 2019 – Dec 2023*

Research Experience

MCADS Lab, Khoury College of Computer Sciences, Northeastern University — Boston, MA
Student Researcher — Egocentric Vision, advised by Prof. Ehsan Elhamifar — *Jun 2026 – Present*

- Designing a dual-egocentric video benchmark for **per-hand ownership attribution** during cooperative tasks: given a hand visible in an egocentric stream, decide whether it belongs to the camera wearer or to the partner, in close-up settings where body and arm cues are out of frame
- Built the capture stack on two Meta Project Aria Gen 1 glasses, using TICSync for cross-device time alignment and Aria MPS multi-SLAM for co-registration of the two wearers into a shared coordinate frame; validated recording profiles, SLAM camera streams, and thermal stability before full-scale collection
- Designed ground truth from dual-egocentric mutual observation, where each wearer's camera supplies the partner-view label for the other, with optional exocentric cameras for offline validation; annotation pipeline combines SAM 3 hand masks with WiLoR 3D hand pose
- Structured the task suite along three axes — physical coupling, coordination structure, and resource contention — and built the evaluation around contrastive minimal pairs, reporting *swap rate* as the headline metric alongside cross-view perspective-flip probes
- **Status:** pipeline validated; data collection in progress, arXiv preprint with baseline evaluations planned for winter 2026–27

Selected Projects

AlphaZero-Style Gomoku Agent | *Python, PyTorch* — Mar 2026

- Implemented an AlphaZero-inspired agent combining Monte Carlo Tree Search with a dual-headed policy–value CNN trained from self-play, with the game environment (legal-move masking, terminal detection) and a pure-MCTS baseline built for controlled comparison

Drug Toxicity Prediction (LD50) | *Python, PyTorch, Scikit-learn, RDKit* — Mar 2026

- Built regression models for acute toxicity prediction on the LD50_Zhu benchmark (Therapeutics Data Commons) from ECFP4 Morgan fingerprints
- Compared linear, random forest, and deep baselines under random and scaffold splits, quantifying the generalization gap that random-split benchmarking hides

Neural Architecture Comparison for Molecular Property Prediction | *Python, PyTorch Geometric* — Nov 2025

- Implemented and compared five architectures (DNN, CNN, RNN/LSTM, Transformer, GNN) for drug classification from SMILES; graph networks reached 85% accuracy, 12% over the DNN baseline under matched tuning and cross-validation

Technical Skills

Languages: Python, C++, C, Java, R, JavaScript/TypeScript, Bash
Vision & ML: PyTorch, PyTorch Geometric, TensorFlow, Scikit-learn, NumPy, OpenCV, SAM 3, WiLoR, MediaPipe
Egocentric Capture: Project Aria SDK, Aria MPS multi-SLAM, TICSync device sync, VRS, camera calibration
Computing: Slurm / H200 GPUs (Northeastern Explorer), Git, Docker, Conda, LaTeX, Next.js, React, Flask